Configuration & Setup Guide

This document explains the parameters in configuration.json and the setup process for the Asteroid Photometry Pipeline.

1. configuration.json Parameters

This file controls the simulation's behavior. Below is a detailed breakdown of each key:

Parameter	Description
asteroid_name	The name of your asteroid (must match the .obj file located in the Shapes/ folder).
blender_path	Absolute path to blender.exe. If left empty, the pipeline will automatically search common installation directories.
JPL_ID	The official identifier used to retrieve ephemerides from JPL Horizons.
USE_DIRECT_RA_DEC	true to use Right Ascension and Declination directly; false to use Ecliptic Lambda/Beta coordinates.
AUTO_CALIBRATE_OFFSET	true to automatically shift the rotation phase (W0) to match DAMIT reference data.
PLOT_BOTH_CURVES	true to generate a comparison plot between your synthetic lightcurve and DAMIT data.
starting_date	The start date and time of the simulation (YYYY-MM-DD HH:MM:SS).
finish_date	The end date and time of the simulation (YYYY-MM-DD HH:MM:SS).
timestep_minutes	The time interval in minutes between each rendered frame (10 minutes minimum).
spin_ra / spin_dec	Coordinates of the asteroid's spin axis in degrees.
W0	Initial rotation phase at JD0 (not always JD2000).

Parameter	Description
spin_lambda / spin_beta	Ecliptic coordinates for the spin axis (used if USE_DIRECT_RA_DEC is false).
phi0	Initial rotational angle in the ecliptic frame.
JD0	The Julian Date corresponding to W0 or phi0.
camera_orthoscale	The scale of the camera orthographic view (adjust this to fit your 3D model properly inside the frame).
P_hours	The rotation period of the asteroid in hours.

2. Blender Setup Process

The pipeline uses Codes/Setup_blender.py to automatically prepare your 3D environment inside cache/object.blend before rendering.

- **What it does:** It cleans the scene, imports your .obj model from Shapes/, centers it, fixes the normals, applies smooth shading, and sets up the Sun and Orthographic Camera.

Customizing Blender Setup Parameters

If you want to tweak the visual appearance or lighting of your renders, you can open Codes/Setup_blender.py and modify these key parameters:

- **Material Properties (Principled BSDF):**
    - Base Color: Changes the base color/albedo of the asteroid surface (default is dark grey 0.18).
    - Roughness: Controls how matte or shiny the asteroid surface appears (higher values mean a rougher, more scattered reflection).
    - Metallic: Kept at 0 for rocky bodies, but adjustable if modeling metallic asteroids.
- **Sun Light Properties:**
    - sun.data.energy: Sets the base energy/intensity of the sun light (default is 10, adjusted dynamically per frame based on solar distance).
    - sun.data.angle: Defines the base angular size of the sun at 1 AU (default is 0.00925 radians, simulating the real Sun's angular diameter). *Note: During rendering, this value also scales dynamically inverse to the solar distance to maintain correct shadow softness as the asteroid moves.*

- **Camera Settings:**
    - cam.data.ortho_scale: Controls the zoom/orthographic scale of the camera (passed from your configuration.json) to properly frame your asteroid model.